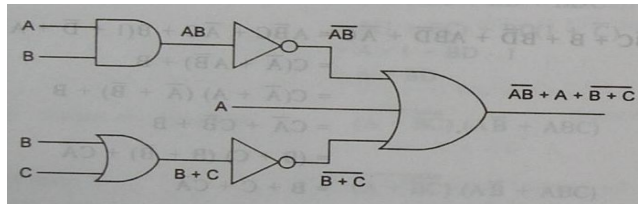
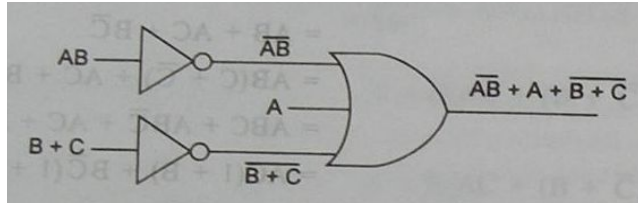
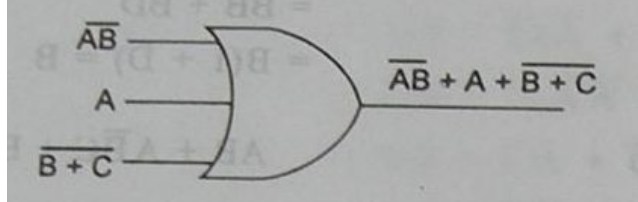


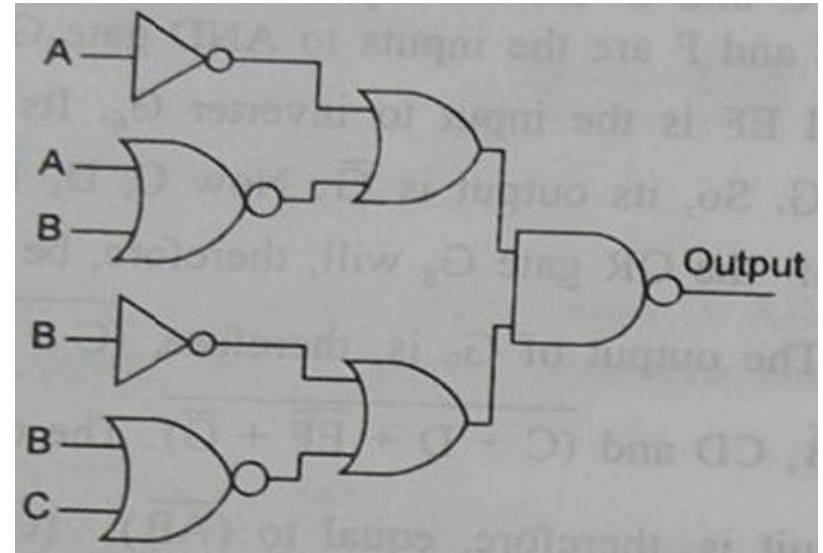
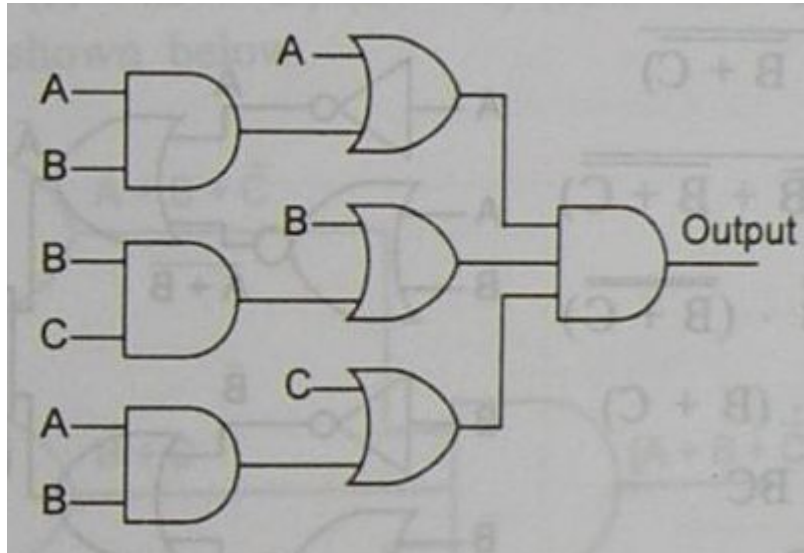
LOGIC IMPLEMENTATION

Boolean Expression to Logic

$$F = (AB)' + A + (B+C)'$$

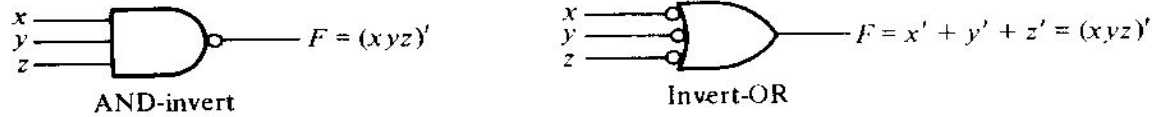


Logic to Boolean Expression

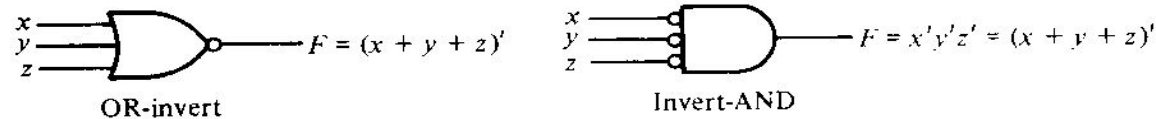


NAND & NOR IMPLEMENTATION

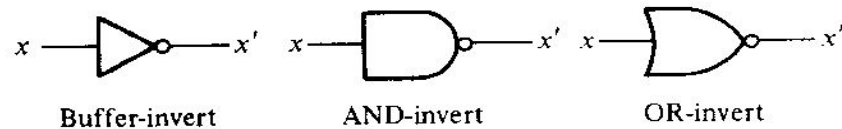
- Easier to fabricate with electronic components
- They are basic gates used in all IC digital logic families



(a) Two graphic symbols for NAND gate.



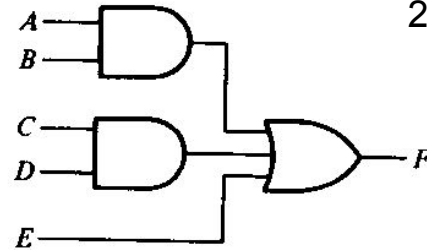
(b) Two graphic symbols for NOR gate.



(c) Three graphic symbols for inverter.

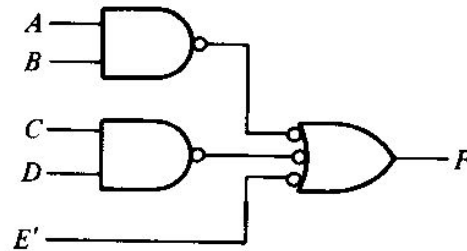
NAND IMPLEMENTATION

$$F = AB + CD + E$$

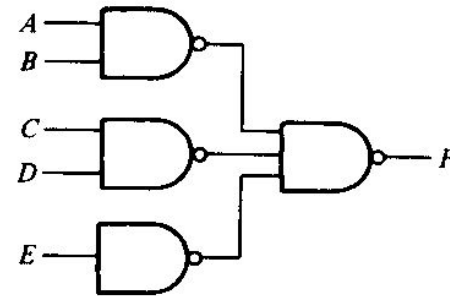


2-level AND-OR = 2-level NAND-NAND

(a) AND-OR



(b) NAND-NAND

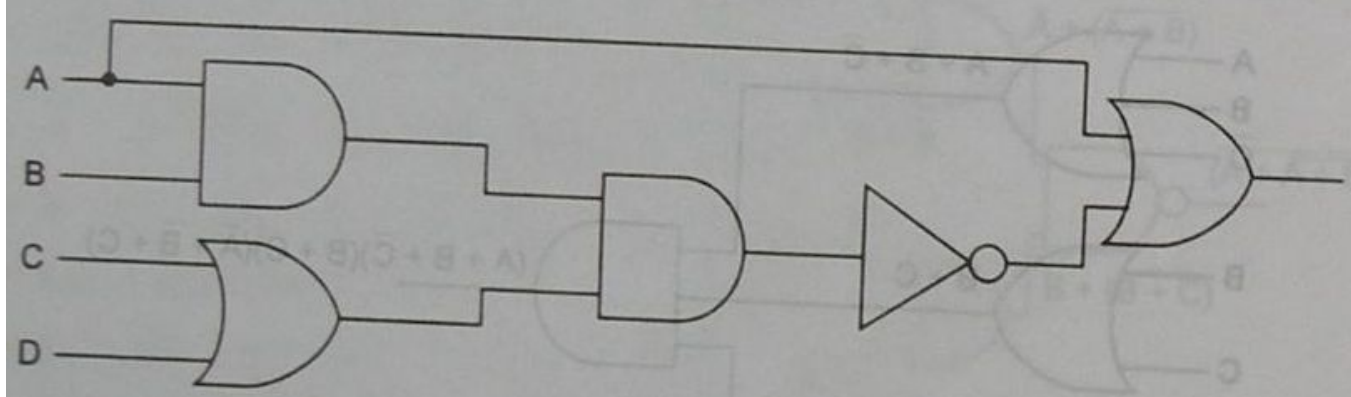


(c) NAND-NAND

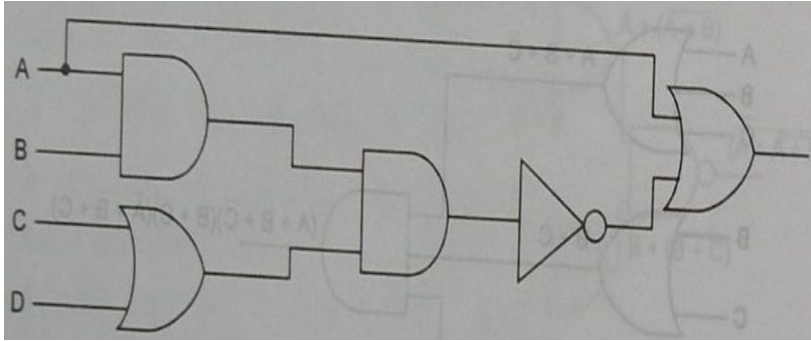
Steps

- Simplify function and express it in **Sum of Products (SOP)** form
- Draw the circuit in AOI logic
- Add a circle at the output of each AND gate and at the inputs to all the OR gates
- Add or subtract an inverter on each line that received a circle in step 3 so that polarity of signals on those lines remains unchanged from that of the original diagram

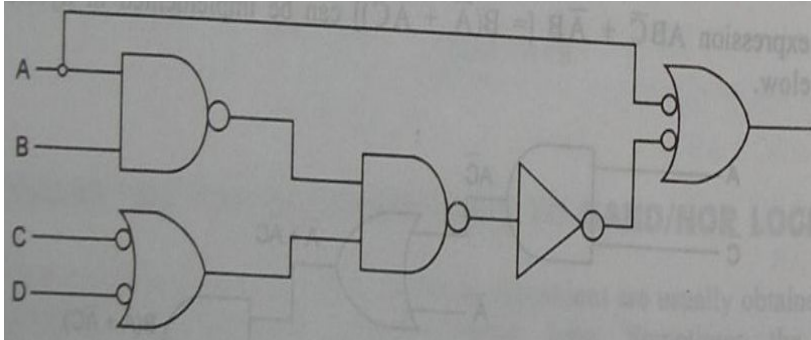
Convert AOI logic circuit to NAND logic



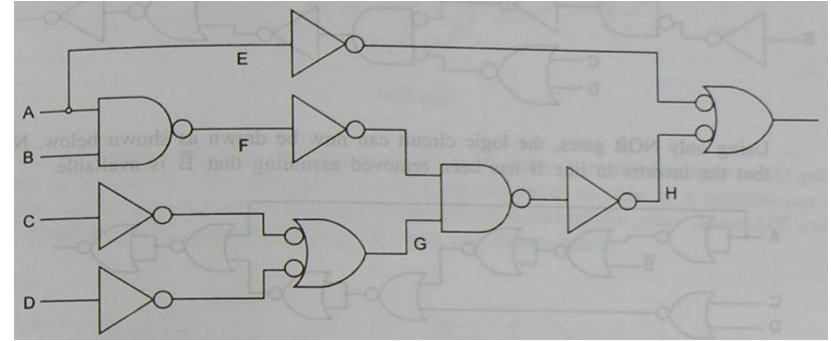
Example



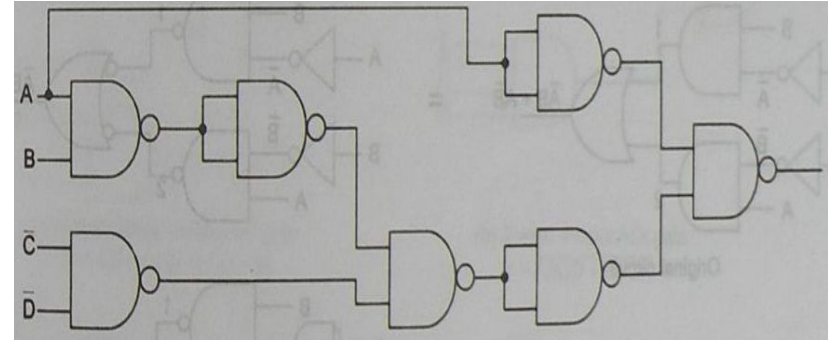
Step 1



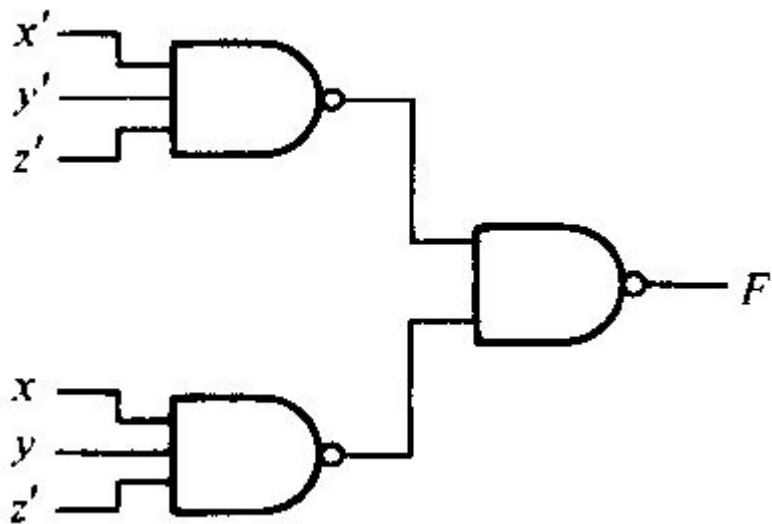
Step 2



Step 3



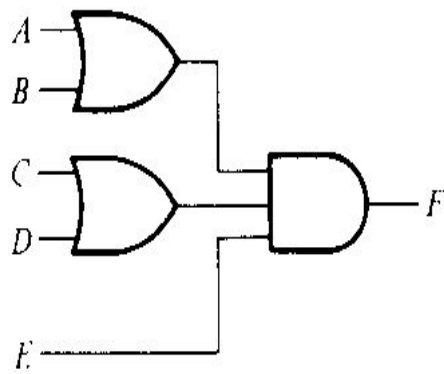
$$F = X'Y'Z' + XYZ'$$



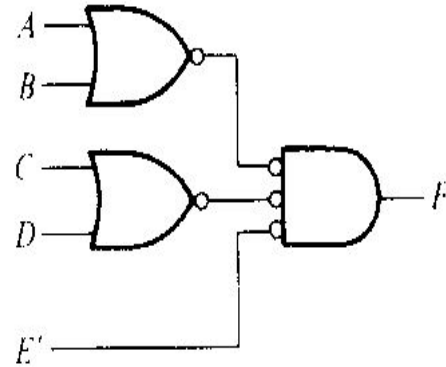
NOR IMPLEMENTATION

$$F = (A+B)(C+D)E$$

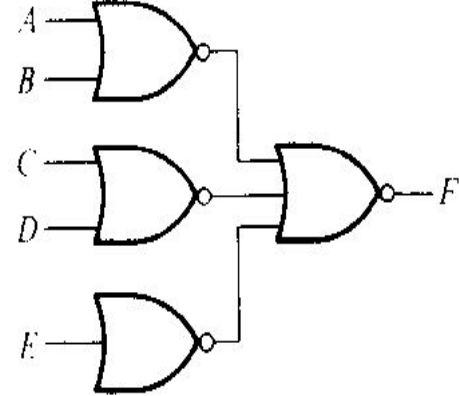
2-level OR-AND = 2-level NOR-NOR



(a) OR-AND



(b) NOR-NOR

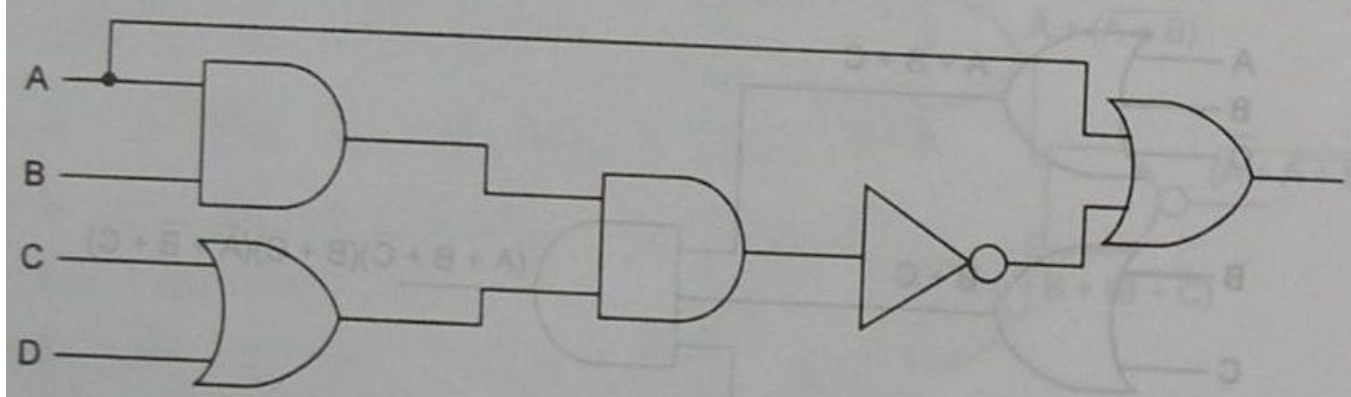


(c) NOR-NOR

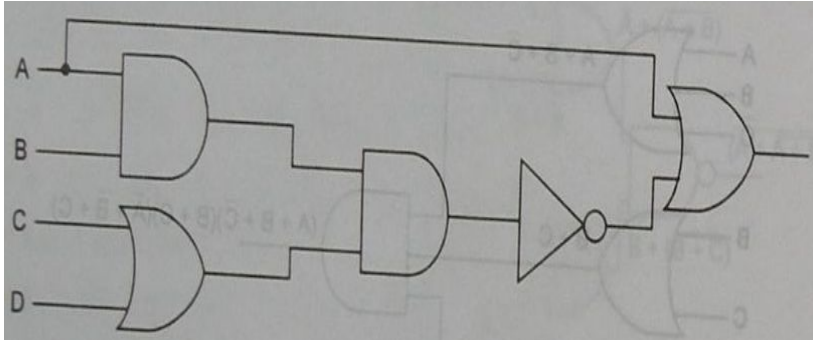
Steps

- Simplify function and express it in **Product of Sums (POS)** form
- Draw the circuit in AOI logic
- Add a circle at the output of each OR gate and at the inputs to all the AND gates
- Add or subtract an inverter on each line that received a circle in step 3 so that polarity of signals on those lines remains unchanged from that of the original diagram

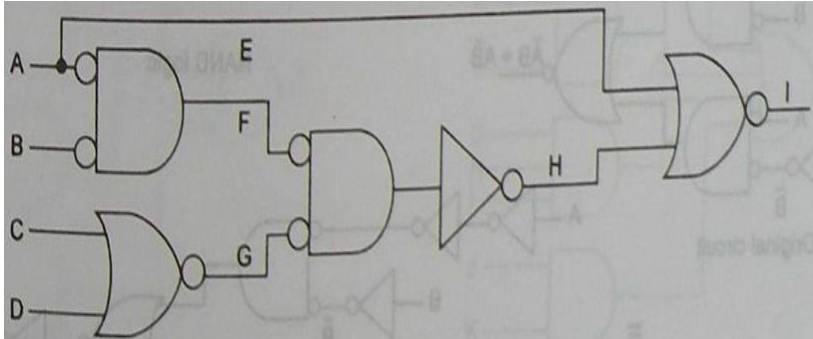
Convert AOI logic circuit to NOR logic



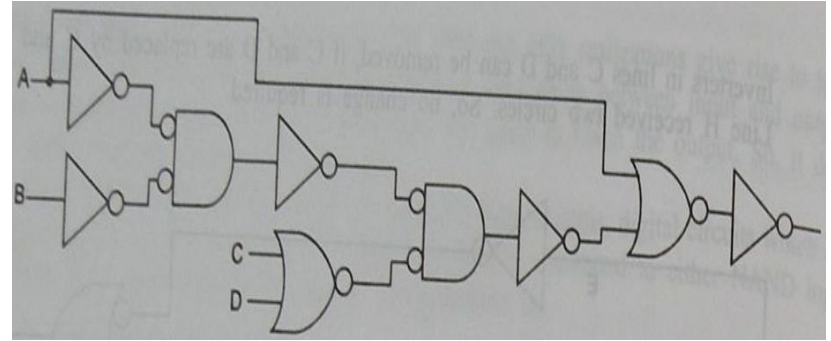
Example



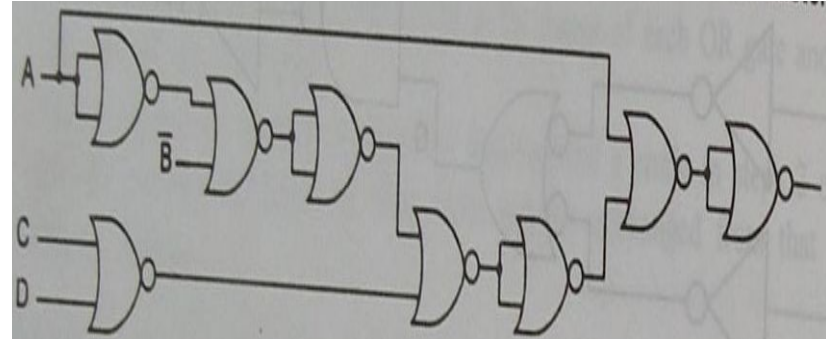
Step 1



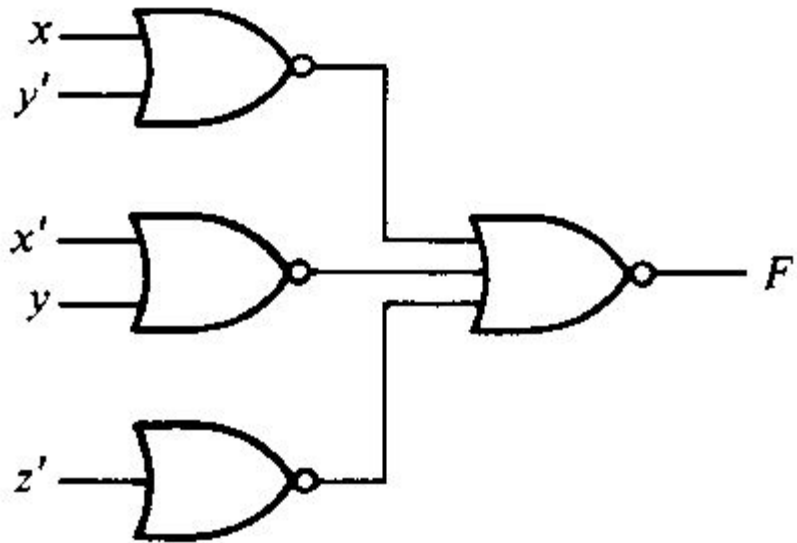
Step 2



Step 3



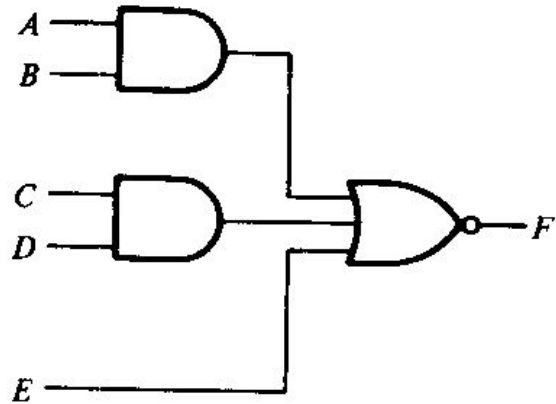
$$F = (X+Y')(X'+Y)Z'$$



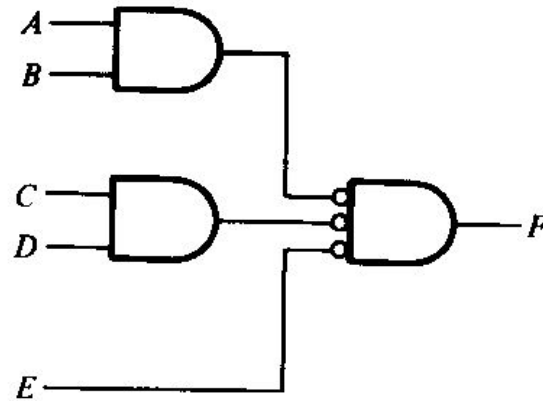
AND-OR-INVERT IMPLEMENTATION

NAND-AND and AND-NOR are equivalent

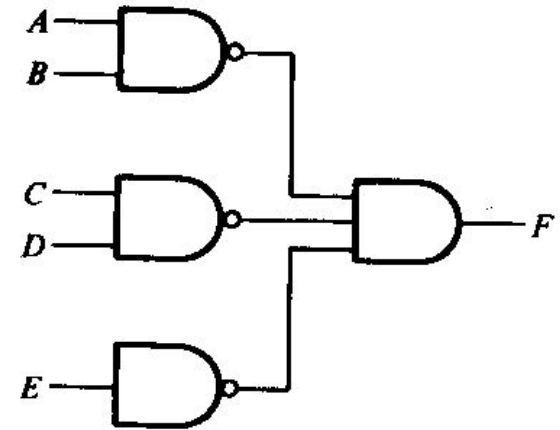
Example: $(AB+CD+E)'$



(a) AND-NOR



(b) AND-NOR,

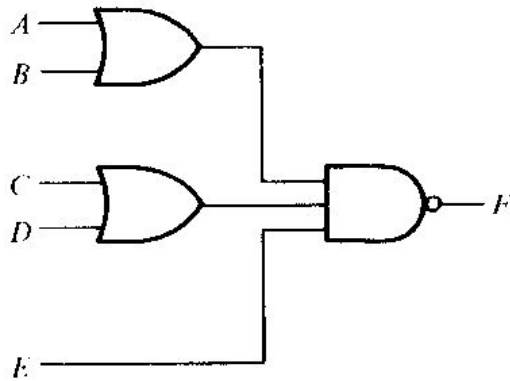


(c) NAND-AND

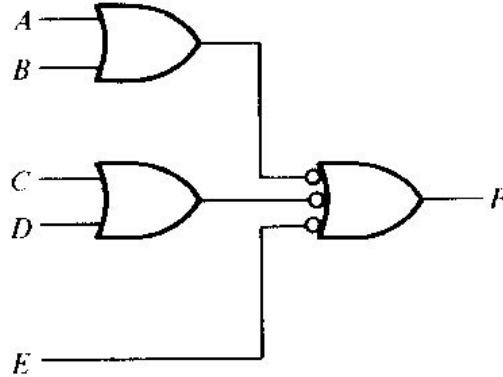
OR-AND-INVERT IMPLEMENTATION

OR-NAND and NOR-OR are equivalent

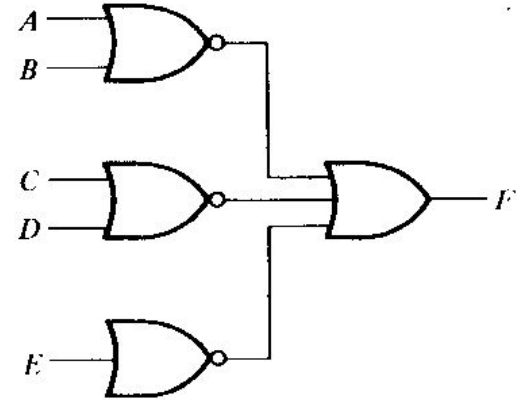
Example: $F = [(A+B)(C+D)E]'$



(a) OR-NAND



(b) OR-NAND



(c) NOR-OR

Degenerate Forms

- Number of levels can be reduced

AND-AND

AND-NAND

NAND-OR

NAND-NOR

OR-OR

OR-NOR

NOR-AND

NOR-NAND

Non-Degenerate Forms

- Number of levels cannot be reduced

AND-OR

NAND-NAND

NOR-OR

OR-NAND

OR-AND

NOR-NOR

NAND-AND

AND-NOR

Active Low Notation

